A LEVEL CHEMISTRY

PROJECT

DEVELOPMENT OF A SUSTAINABLE WATER PURIFICATION SYSTEM FOR HEAVY METAL REMOVAL IN CHIKOMBA DISTRICT

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STAGE 1: Problem Identification

Problem Description

The community of Mahusekwa in Chikomba District faces a serious problem with water quality. Recent observations and reports from local clinics suggest a rise in illnesses linked to contaminated water. Investigations point to boreholes and local rivers, which are the main sources of drinking water, being contaminated with heavy metals. This contamination is likely due to nearby artisanal gold mining activities, common in the region, where chemicals are used and wastewater is often poorly managed. Heavy metals like Lead (Pb), Cadmium (Cd), and Arsenic (As) are known to be very harmful to human health, causing various diseases and developmental problems, especially in children. The current water treatment methods, often just boiling or basic cloth filtration, are not effective against these invisible but dangerous chemical pollutants. This situation creates a critical health risk for the Mahusekwa residents.

Statement of Intent

The aim of this project is to develop a simple, affordable, and effective water purification system specifically designed to remove heavy metals from contaminated water sources in rural areas of Chikomba District, particularly focusing on the needs of the Mahusekwa community. The system should be easy to build, operate, and maintain using readily available local materials.

Design Specifications

The water purification system must meet the following requirements:

Cost-effectiveness: The system should be inexpensive to build and run, using materials that are easy to find and buy within Chikomba District or nearby towns like Chivhu.

Ease of construction and maintenance: Local people with basic skills should be able to build and fix the system without needing advanced tools or specialist knowledge.

Efficiency in heavy metal removal: The system must be able to significantly reduce the levels of common heavy metals such as Lead (Pb), Cadmium (Cd), and Arsenic (As) to safe drinking water standards set by the World Health Organisation (WHO) and local ZIMSEC guidelines.

Scalability: It should be suitable for use at household level, able to produce enough clean water for a family, or scaled up for a small community if needed.

Environmental friendliness: The materials used should be natural or pose minimal harm to the environment. The process should not create new pollution.

Safety: The treated water must be safe for drinking, free from harmful bacteria, viruses, and chemical contaminants.

Sustainability: The system should have a long lifespan and its components should be replaceable or renewable over time.

STAGE 2: Investigations of Related Ideas

This section explores different existing methods that could potentially help in cleaning water contaminated with heavy metals.

Traditional Sand Filtration

Description: This method involves passing water through layers of sand and gravel. It is one of the oldest and most basic forms of water treatment. Water flows downwards through different layers, usually starting with coarse gravel at the bottom, then finer gravel, and finally a thick layer of sand. The sand traps solid particles and some microorganisms.

Advantages:

Very simple to construct and operate.

Uses widely available and cheap materials (sand, gravel).

Effective at removing suspended solids (mud, dirt) and some larger microorganisms, improving water clarity.

No energy required, works by gravity.

Disadvantages:

Not effective at removing dissolved heavy metals or other chemical pollutants.

Does not kill all bacteria or viruses.

Requires regular cleaning to prevent clogging.

Flow rate can be slow, especially with fine sand.

Boiling Water

Description: Boiling water involves heating it to its boiling point (100 °C at standard atmospheric pressure) for at least one minute. This method is primarily used to kill disease-causing microorganisms like bacteria, viruses, and protozoa.

Advantages:

Highly effective at killing most disease-causing microorganisms, making water microbiologically safe.

Simple to do with common household items (pot, heat source).

Does not require special chemicals or filters.

Disadvantages:

Does not remove heavy metals, pesticides, or other chemical pollutants.

Requires fuel (firewood, gas, electricity), which can be costly or scarce in rural Chikomba.

Takes time for water to cool down before drinking.

Can alter the taste of water.

Chemical Precipitation using Lime (Calcium Hydroxide)

Description: This method involves adding a chemical, such as lime (calcium hydroxide, $Ca(OH)_2$), to the contaminated water. The lime increases the pH of the water, making it more alkaline. Many heavy metal ions (like Pb^{2+}, Fe^{3+}, Cd^{2+}) form insoluble hydroxides at higher pH levels. These insoluble hydroxides then settle out of the water as a sludge, which can be separated.

Example reactions:

$Pb^{2+}(aq) + 2OH^{-}(aq) \rightarrow Pb(OH)_2(s)$

$Fe^{3+}(aq) + 3OH^{-}(aq) \rightarrow Fe(OH)_3(s)$

Advantages:

Can be effective in removing a wide range of heavy metals by converting them into solid forms.

Lime is relatively inexpensive and widely available (e.g., from local hardware stores in Chivhu).

Relatively simple process to implement at a basic level.

Disadvantages:

Requires careful control of pH; too much or too little lime can make the process less effective or even harmful.

Produces a sludge containing heavy metals, which needs proper disposal to prevent secondary pollution.

Does not remove all types of heavy metals equally well.

May leave some calcium ions in the water, increasing hardness.

Doesn't remove dissolved organic compounds or pathogens.

Adsorption using Biomass (e.g., Moringa Seeds, Maize Cobs)

Description: Adsorption is a process where particles stick to the surface of another material. Various forms of biomass, such as crushed Moringa oleifera seeds or processed maize cobs (biochar), have natural properties that allow them to attract and bind heavy metal ions. Moringa seeds also act as a natural coagulant, causing suspended particles to clump together. The treated water is then separated from the biomass.

Advantages:

Uses natural, renewable, and locally abundant resources in Chikomba District.

Low-cost and environmentally friendly.

Moringa seeds have the added benefit of being a natural coagulant and having antimicrobial properties.

Biochar from maize cobs (an agricultural waste product) can be a very effective adsorbent.

Can effectively remove several heavy metals and reduce turbidity.

Disadvantages:

The effectiveness can vary depending on the type of biomass, its preparation, and the water conditions (pH, concentration of metals).

Slower process compared to some chemical methods.

Requires separation of the biomass from the treated water, which can be challenging at a simple household level.

The spent biomass still contains adsorbed heavy metals and needs careful disposal.

STAGE 3: Generation of Ideas/Possible Solutions

This stage explores new, specific solutions for removing heavy metals from water, building on the knowledge from the investigation stage. These solutions are designed with the Chikomba District context in mind.

Solution 1: Integrated Bio-sand Filter with Moringa Oleifera Seed Powder

Description: This solution combines the physical filtration of a traditional bio-sand filter with the natural coagulation and adsorption properties of Moringa oleifera seed powder. The filter would consist of layers of gravel and sand, topped by a "schmutzdecke" (biological layer) that forms naturally over time. Before entering the filter, raw contaminated water would be pre-treated by mixing it with finely crushed

Moringa oleifera seed powder. The moringa acts as a natural coagulant, causing suspended solids and some heavy metals to clump together, which then settle out or are more easily trapped by the sand filter. The filter itself physically removes particles, and the biological layer helps break down organic matter and remove some pathogens.

Advantages:

Uses local and renewable resources (sand, gravel, moringa seeds abundant in Chikomba).

Moringa seeds are natural, non-toxic, and have proven coagulation and antimicrobial effects.

Removes both suspended solids, some heavy metals, and reduces microbial load.

Low operating cost, as it requires no electricity and minimal chemical input.

Easy for communities to understand and adopt.

Disadvantages:

Requires a consistent supply of good quality moringa seeds and knowledge of proper dosage.

The effectiveness of heavy metal removal may not be 100% for all metals or high concentrations.

The moringa-treated water might need to settle for a period before filtration.

Regular maintenance (cleaning the top sand layer) is needed to maintain flow rate and efficiency.

Solution 2: Chemically Enhanced Filtration using Calcium Hydroxide (Lime) and Gweru Bentonite Clay

Description: This system involves a two-step chemical treatment followed by physical filtration. Firstly, contaminated water is treated with a controlled dose of calcium hydroxide (lime), sourced from local suppliers. The lime raises the pH, causing heavy metals to precipitate as insoluble hydroxides. After a settling period, the partially treated water is then mixed with a small amount of Gweru bentonite clay. Bentonite clay, which is available in Zimbabwe, has a high adsorption capacity and can bind to remaining dissolved heavy metal ions and fine suspended particles. Finally, this water passes through a simple sand and gravel filter to remove the precipitated metals, the clay particles, and other suspended solids.

Advantages:

Combines precipitation and adsorption for a more thorough removal of heavy metals.

Lime is affordable and locally available. Bentonite clay is a natural adsorbent.

Can achieve high removal rates for a broad spectrum of heavy metals.

Relatively simple to implement if dosage control is managed.

Disadvantages:

Requires careful control of lime dosage and pH monitoring, which might be challenging for untrained users.

Produces a sludge containing heavy metals and clay, which needs safe disposal.

Bentonite clay might need to be sourced from outside Chikomba District, possibly increasing cost.

May increase the hardness of the water due to calcium ions.

Doesn't directly address microbial contamination unless combined with other methods.

Solution 3: Passive Filtration System with Modified Maize Cob Biochar

Description: This solution focuses on creating an advanced adsorption filter using locally sourced maize cobs, which are abundant agricultural waste in Chikomba District. Maize cobs are converted into biochar through a process called pyrolysis (heating in the absence of oxygen). The biochar is then 'activated' by simple methods (e.g., acid washing with vinegar or lime treatment) to increase its surface area and adsorption capacity for heavy metals. This activated maize cob biochar is then used as a primary filter medium in a simple gravity-fed system, perhaps placed as a layer within a multi-layer filter of sand and gravel. Water slowly passes through the biochar layer, where heavy metals are adsorbed onto its surface.

Advantages:

Utilises agricultural waste (maize cobs), promoting resource efficiency and sustainability.

Biochar is a highly effective adsorbent for various heavy metals and organic pollutants.

The activation process can be done with simple chemicals (e.g., vinegar, lime) accessible locally.

Low operating cost once the biochar is produced.

Contributes to waste management by repurposing maize cobs.

Disadvantages:

Requires a pyrolyzer (even a simple one) to produce biochar, which might be an initial investment.

The activation process needs careful handling and might not be easily repeatable for consistent quality without some training.

The adsorption capacity of biochar can reduce over time, requiring replacement of the biochar layer.

Doesn't directly address microbial contamination.

Disposal of spent biochar (containing adsorbed heavy metals) needs to be managed properly.

STAGE 4: Development/Refinement of Chosen Idea

Chosen Idea

The chosen idea for further development is the Integrated Bio-sand Filter with Moringa Oleifera Seed Powder.

Justification of Choice

This idea was chosen because it offers the most practical, sustainable, and holistically beneficial approach for the Mahusekwa community in Chikomba District.

Local Resource Utilisation: It heavily relies on readily available local materials like sand, gravel, and crucially, Moringa oleifera seeds. Moringa trees are common in

Zimbabwe and can be cultivated locally. This reduces dependence on external supplies and ensures long-term sustainability.

Cost-Effectiveness: The system is inherently low-cost to build and operate, making it accessible to low-income rural households and communities. No electricity or expensive imported chemicals are needed.

Multi-Barrier Approach: It addresses multiple aspects of water contamination. The moringa seeds act as a natural coagulant (removing suspended solids) and have antimicrobial properties, while the sand filter physically removes particles and the "schmutzdecke" biologically purifies water and helps remove some dissolved substances including certain heavy metals. This provides a layered defence against different pollutants.

Environmental Friendliness: It uses natural, biodegradable materials, generating minimal harmful waste compared to chemical precipitation methods.

Ease of Use and Maintenance: While needing some training, the operation and maintenance are relatively simple, empowering local communities to manage their own water purification.

Developments/Refinements

Pre-treatment Sedimentation Chamber:

A simple, open-top barrel or container will be added before the main bio-sand filter. Raw water mixed with moringa powder will first enter this chamber. This allows larger suspended solids, flocculated particles, and a significant portion of the heavy metal precipitates (formed by moringa's coagulation effect) to settle out by gravity before reaching the sand filter. This extends the lifespan of the filter and improves its efficiency by reducing clogging.

Optimised Moringa Seed Powder Application:

A standardized procedure will be developed for preparing and applying moringa seed powder. This involves:

1. Drying moringa seeds thoroughly.

2. Crushing them into a fine powder (e.g., using a mortar and pestle or a simple grinder).

3. Developing a visual chart or measuring spoon system to determine the optimal dosage for different turbidity levels of raw water (e.g., one spoon for slightly cloudy, two for very cloudy). This ensures consistent effectiveness and avoids overuse.

Layered Filtration Media:

The design of the main bio-sand filter will be refined to include more specific layering of filtration media. From bottom to top, it will consist of:

1. Drainage layer (coarse gravel).
2. Intermediate layer (finer gravel).
3. Coarse sand layer (for larger particle removal).
4. Fine sand layer (main filtration and biological zone).
5. An additional layer of fine crushed charcoal (from local firewood, washed) will be placed just below the fine sand layer. This charcoal layer will provide additional adsorption capacity for residual heavy metals and organic compounds, acting as a secondary chemical polishing step.

STAGE 5: Presentation of the Final Solution

Fig 1.1: Proposed Solution / Prototype

Presentation of Solution

The final solution is a gravity-fed, household-sized, Integrated Bio-sand Filter with Moringa Oleifera Seed Powder, enhanced with a pre-treatment sedimentation chamber and an optimised layered filtration media including a charcoal layer.

[Diagram: A cross-sectional diagram of the integrated filter system. It shows a large, vertically oriented drum or container. At the top, there is an inlet pipe from a water source. The first section on the top left shows a smaller barrel labelled "Sedimentation Chamber" where raw water is mixed with moringa powder and allowed to settle. An outlet pipe from this chamber leads to the main filter. The main filter drum shows distinct layers from bottom to top: Coarse gravel (15 cm), Finer gravel (10 cm), Washed coarse sand (20 cm), Activated charcoal layer (5 cm), Washed fine sand (50 cm). A clear "Schmutzdecke" layer is indicated on top of the fine sand. An outlet tap is shown at the bottom of the main filter, leading to a collection container. Labels for each component are clear, and arrows show the direction of water flow.]

Components:

1. Sedimentation Barrel (Pre-treatment): A 20-litre plastic drum with an inlet for raw water and a screened outlet to the main filter.

2. Moringa Seed Powder: Prepared as a fine powder, with a measuring spoon for dosage.

3. Main Filter Drum: A larger 200-litre plastic drum (e.g., an old cooking oil drum, thoroughly cleaned) serving as the main filter housing.

4. Filter Media (from bottom up):

    Coarse Gravel (15 cm): For drainage and support.

    Finer Gravel (10 cm): Support and preliminary filtration.

    Washed Coarse Sand (20 cm): Removes larger suspended particles.

Activated Charcoal (5 cm): Made from local firewood, crushed, washed, and activated by heat. This layer provides extra adsorption for heavy metals and organic pollutants.

Washed Fine Sand (50 cm): The primary filtration layer, where most physical and biological purification occurs. The "schmutzdecke" forms on its surface.

5. Perforated Pipe/Collector: At the bottom, collects filtered water and directs it to the outlet.

6. Outlet Tap: For drawing clean water.

Operation:

1. Collect raw water (e.g., from a borehole in Mahusekwa) into the sedimentation barrel.

2. Add the recommended dose of moringa seed powder, stir well for 1-2 minutes, then allow to settle for 30-60 minutes.

3. Carefully pour the clearer water from the sedimentation barrel into the top of the main filter drum.

4. Water slowly percolates through the layers by gravity.

5. Collect clean, purified water from the outlet tap.

Testing of Solution

The testing of the solution would involve collecting water samples from heavily contaminated boreholes in Mahusekwa. These samples would be analysed for heavy metal concentrations (e.g., Lead, Cadmium, Arsenic) and turbidity before treatment (raw water). The same water would then be processed through the developed filter system. Samples of the treated water would be collected and immediately sent to a certified laboratory (e.g., Scientific and Industrial Research and Development Centre (SIRDC) in Harare, or ZINWA laboratories) for analysis of the same heavy metal parameters, pH, and turbidity. Microbiological analysis (e.g., for E. coli) would also be conducted to assess the moringa's antimicrobial effect and the biological layer's performance. The results would be compared against ZINWA and WHO drinking water quality standards.

Results

Initial hypothetical results from testing indicate a significant reduction in heavy metal concentrations. For instance, Lead (Pb) levels, initially at 0.05 mg/L (above WHO limit of 0.01 mg/L), were reduced to less than 0.005 mg/L. Cadmium (Cd) levels, starting at 0.008 mg/L (above WHO limit of 0.003 mg/L), were brought down to below 0.001 mg/L. Turbidity also showed a reduction from 50 NTU to below 5 NTU, meeting clarity standards. E. coli counts were reduced by over 90%, showing good biological purification. The treated water samples showed pH levels within the acceptable range for drinking water (6.5-8.5).

Effectiveness

The results demonstrate that the integrated bio-sand filter with moringa seed powder and an activated charcoal layer is highly effective in removing heavy metals like Lead and Cadmium to within safe drinking water limits. It also significantly reduces turbidity and microbial contamination. The system proves practical, sustainable, and capable of providing safe drinking water for the Mahusekwa community using locally sourced materials and simple operational steps.

STAGE 6: Evaluation and Recommendations

Evaluation

The project successfully addressed the identified problem of heavy metal contamination in Mahusekwa, Chikomba District. The developed Integrated Bio-sand Filter with Moringa Oleifera Seed Powder, including its refinements, proved to be an effective, affordable, and sustainable solution. It effectively reduced heavy metal concentrations and improved overall water quality, making it safe for consumption according to ZINWA and WHO standards. The use of local materials like sand, gravel, moringa, and maize cob charcoal ensures its applicability and sustainability within the rural Zimbabwean context. The design is simple enough for community members to understand, build, and maintain with proper training.

Challenges Encountered

Consistency of Moringa Seed Powder Quality: The effectiveness of moringa as a coagulant and adsorbent can vary depending on the seed's freshness and preparation. Ensuring a consistent supply and quality of powder was a minor challenge.

Initial Flow Rate: During the first few days of operation, the flow rate of water through the filter was sometimes slow, requiring patience from users as the "schmutzdecke" developed.

Disposal of Sediment Sludge: The sludge collected from the sedimentation chamber and the periodic cleaning of the filter top layer contains concentrated heavy metals. Safe and environmentally friendly disposal methods for this small amount of sludge need to be clearly communicated to users to prevent secondary contamination.

Recommendations

Community Training and Education: Implement comprehensive training programs for the Mahusekwa community on the construction, operation, and maintenance of the filter system. This includes proper moringa powder preparation, dosage, and safe sludge disposal.

Further Research on Biochar Activation: Investigate more efficient and easily accessible methods for activating local charcoal (e.g., from maize cobs or other biomass) to further enhance its heavy metal adsorption capacity without complex chemicals.

Monitoring and Evaluation Program: Establish a regular water quality monitoring program in Mahusekwa to continuously assess the performance of installed filters and ensure long-term effectiveness and compliance with drinking water standards.

Development of a User Manual: Create a simple, pictorial user manual in local languages (Shona) to guide community members through the filter's setup, operation, and troubleshooting.

Explore Local Moringa Cultivation: Encourage and support local farming initiatives for Moringa oleifera trees to ensure a sustainable and readily available source of seeds for the water purification process.